



Speaking through cave art

Cave art is located in acoustic hotspots deep within caves, indicating that the art might represent what was spoken there. <http://dgit.in/cavetalk>

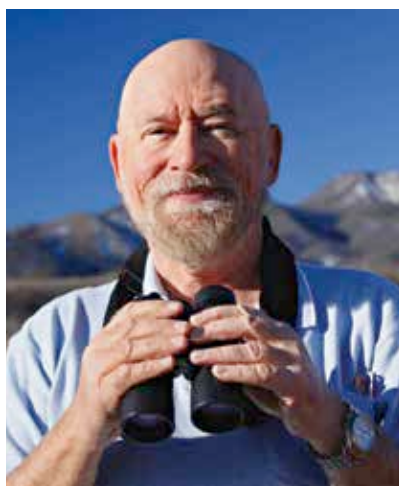


This star should not exist

Researchers have found a star in the Milky Way with an incredibly low amount of metals, challenging current star formation models. <http://dgit.in/wyrdstr>

SO WILL WE BE ABLE TO UNDERSTAND THEM?

One of the main purposes of technology is to solve problems we humans are incapable of, or rather slow at. When it comes to bridging the gap in human animal communication, technology like wearables and AI are coming to the aid of research scientists and inventors.



Dr Con Slobodchikoff

Dr Con Slobodchikoff, an animal behavior expert, has been researching prairie dogs (tiny rodents that live in burrows beneath the plains) and their vocalisation for more than 30 years. His research has shown him that not only do prairie dogs have complex warning calls, they even communicate the colour of an individual's clothing to each other. Last year, Dr. Slobodchikoff and a colleague created an algorithm to parse these vocalisations into English, effectively creating a prairie dog translator and also started Zoolingua, a startup dedicated to decoding animal languages. And he believes that the results he has achieved with prairie dogs can be achieved with other species too.

His efforts involve showing his AI programme thousands of videos of dog behaviors and vocalisations for it to learn from. To start off, these algorithms are going to identify and distinguish dog behavior like tail wagging, a low bark, gritted teeth and more. At this point, they'll

need human intervention to associate meaning with these actions. Over time, with bigger datasets to learn from, the final objective of this effort would be to be able to point your phone camera at a dog that is barking at you, and an app telling you what the dog was saying, what mood it was in, and also highlighting signs of friendliness or danger, based on the dog's posture and body movements, etc.

And it's not just dogs. Dolphins are capable of quite a sophisticated level of communication, pretty close to human language complexity. Swedish language technology company Gavagai AB is working with KTH Royal Institute of Technology in Stockholm to use AI and build a dolphin language dictionary by 2021. But all of these are future goals, what about now?

IN-VEST-ED

While AI and machine learning tech takes its time to help us understand our furry and feathered friends, we have other avenues to help us understand them. For instance, Georgia Institute of Technology researcher Melody Jackson, director of the university's BrainLab, has developed pet communication technology that could save a human life. It doesn't exactly convert their vocals into hu-



Melody Jackson, Associate Professor; Director, Center for BioInterface Research, Georgia Tech

man speech, but teaches them to use a device to do so.

Along with Google Glass developer Thad Starner, Jackson channeled her passion for animals into developing a add-on for a typical dog vest that lets dogs touch their snouts to it in a specific way to play specific, verbal messages. The interesting thing about this is how easily and quickly dogs were able to figure out how to use it. One particular test subject even learnt (within 27 seconds!) that without touching, just hovering his nose over the button can do the trick. It's all part of a field called animal-computer interaction.

A service dog can be trained to perform other actions with the same device, like send an SOS alert with GPS coordinates. The same message can be altered to fulfil a host of purposes. For instance, a guide dog can tell its visually impaired owner exactly what kind of obstacle is there in front of them.

THE FUTURE

In a report co-authored for Amazon (<http://dgit.in/Hearboy>), futurologist William Higham of Next Big Thing says he believes devices that can talk dog could be less than 10 years away. And if the current applications of AI and machine learning to the problem are anything to go by, it is more likely to happen sooner, rather than later.

If the gap in human-animal communication is bridged, innumerable doors would open up. While most of them would be good – like your dog telling you exactly what it is afraid of, helping you help it at the right point – some of it can upset existing establishments as well. For instance, if the same tech is applied to dairy and poultry livestock – would you really be comfortable consuming products from a being that can tell you exactly how much it hates being used for your food? To solve these problems is exactly why we've evolved over all these years to be the most superior species on the face of the Earth. 